**Manual Chapter** : Configuring the BIG-IP System for DHCP Renewal

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# Configuring the BIG-IP System for DHCP Renewal

## Overview: Renewing IP addresses for DHCP clients

You can configure the BIG-IP system to manage DHCP renewal requests and responses.

Before configuring the BIG-IP system to manage DHCP renewal requests and responses, it is helpful to understand some BIG-IP system terminology:

| BIG-IP object type | Definition |
|---|---|
| BIG-IP pool member | A DHCP relay target (such as a DHCP server or BOOTP server). This is the dynamic address server to which the BIG-IP system forwards unicast requests. |
| BIG-IP virtual server | A BIG-IP system address on the listening VLAN |
| BIG-IP VLAN assigned to a virtual server | A listening VLAN, controlled on a per-virtual server basis |

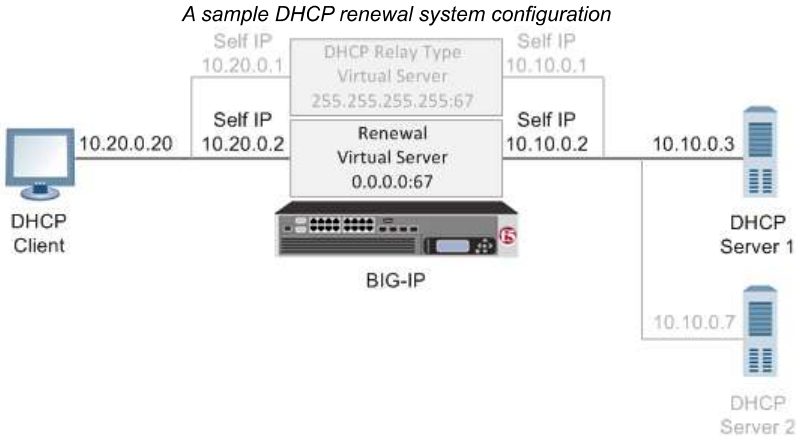
### About DHCP renewal

You can configure the BIG-IP system to act as a DHCP renewal system. A common reason to configure the BIG-IP system as a renewal system is when the DHCP servers reside on a different subnet than that of the client systems, and the BIG-IP system is also configured as a DHCP relay agent. As a DHCP renewal system, the BIG-IP system manages the renewal of client IP addresses by DHCP servers before the addresses expire.

During the renewal process, a DHCP client sends a renewal request, which is passed through a BIG-IP Forwarding IP type of virtual server directly to the specific DHCP server that issued the initial client IP address. The DHCP server then sends a response to renew the lease for the client's IP address.

In the example shown in the illustration, a DHCP client sends a renewal message to the same BIG-IP system that initially acted as the DHCP relay agent. This renewal request is forwarded through a BIG-IP renewal virtual server directly to DHCP server 1. DHCP server 1 then provides a response to renew the lease for the client's IP address.

*A sample DHCP renewal system configuration*



## Creating a DHCP renewal virtual server

A Dynamic Host Configuration Protocol (DHCP) renewal virtual server forwards a DHCP request message from a DHCP client directly to a DHCP server, to automatically renew an IP address before it expires.

1. On the Main tab, click **Local Traffic › Virtual Servers** .
   The Virtual Server List screen opens.
2. Click **Create**.
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. Type a description for the virtual server.
5. From the **Type** list, select **Forwarding (IP)**.
6. For a host, in the **Destination Address/Mask** field, type an IPv4 or IPv6 address in CIDR format to allow all traffic to be translated.

   The supported format is address/prefix, where the prefix length is in bits. For example, an IPv4 address/prefix is `0.0.0.0/0`, and an IPv6 address/prefix is `::/0`.
7. In the **Service Port** field, type `67` (IPv4) or `547` (IPv6).
8. From the **Protocol** list, select **UDP**.
9. From the **VLAN and Tunnel Traffic** list, select the VLANs on the same network as the DHCP clients.
10. Click **Finished**.

The BIG-IP system is now configured with a virtual server that can forward DHCP renewal requests directly to the appropriate DHCP server.

## Implementation result

The BIG-IP system is configured to forward DHCP client renewal requests to appropriate DHCP servers that reside on a different subnet than the client systems. The BIG-IP also forwards the DHCP server responses back to the client systems, therefore ensuring that client IP addresses do not expire.

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